Roll Back the Fault, Not the Future: Causal Recovery for Coding Agents with AGENTTX

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Abstract

Coding agents increasingly execute long, stateful trajectories in which later tool calls reuse filesystem state produced by earlier editing, compilation, and testing. Existing mechanisms couple the lifetime of speculative state to the granularity of recovery: command-scoped isolation repeatedly reconstructs state, whereas coarse temporal rollback discards independent later work. Recovery may target a historical state that never existed as a temporal prefix, and the workspace and agent context must move to that target together. AGENTTX keeps the trajectory in one speculative workspace but recovers only the failed step and its dependents. It captures dependencies and object versions, then uses one recovery decision to rebuild both the workspace and agent-visible context. In the 144-run controlled recovery suite, AGENTTX retained 100% of independent work and removed 100% of invalid descendants; temporal recovery retained only 41% of independent work at 64 calls. At 48 document entries, preserving valid later work avoided 1,425 replay tokens relative to temporal recovery and 3,340 relative to whole-session abort.

1 Introduction

Coding agents increasingly solve software-engineering tasks through long sequences of interactions with a development environment rather than producing a patch in a single turn. Their control loops interleave model reasoning with tool actions and observations, allowing plans to change after commands fail or reveal new repository evidence [1]. Systems such as SWE-agent and OpenHands expose repository navigation, editing, shell commands, and tests to the model [2,3]. SWE-bench evaluates issue resolution across multiple files and test runs [4]. The filesystem therefore becomes persistent execution state. Later steps consume files and artifacts produced earlier in the same trajectory.

Existing systems preserve this evolving state at different boundaries. Command-level semisolates such as `try` stage one opaque invocation [5], whereas agent-facing filesystems and branch runtimes preserve longer-lived workspace state [6,7]. Checkpoints restore temporal state [8,9]; process checkpointing provides restart guarantees [10, 11]. These mechanisms usually recover the state unit they preserve: a command, session, branch, checkpoint, or transaction. Short units repeatedly reconstruct execution state; coarse units can remove valid work together with corrupted effects.

Correct recovery exposes this limitation. Leaving all state in place preserves progress, while removing only the failed operation can leave derived effects behind. Coarse recovery removes those effects, yet also discards unrelated work enclosed by its boundary. To isolate granularity, our study gives coarse baselines exact failure locations and their most favorable native boundaries. Even under these conditions, correct recovery sacrifices independent progress within a turn or checkpoint.

The underlying problem is that execution order does not encode failure dependence. Consider four interleaved steps: s_0 writes a faulty formatter, s_1 uses it to generate a report, s_2 edits an unrelated document, and s_3 tests the faulty output. Recovery should remove $\{s_0, s_1, s_3\}$ while retaining s_2. Undoing only s_0 leaves derived state behind, while restoring a prefix also discards s_2. Changing among turn-level checkpoints, temporal suffixes, and whole branches merely moves this collateral loss across workloads. No fixed coarse boundary consistently avoids it.

Selective recovery raises three systems challenges. First, *dependency discovery* must uncover producer–consumer relations hidden behind shells, compilers, tests, and descendant processes. Second, *object identity* must relate effects across aliases, renames, replacements, and layered filesystems. Third, *selective reconstruction* must rebuild a non-prefix workspace while aligning agent-visible context with that state.

The root cause is a mismatch between how the compared substrates organize recovery state and how failures propagate. Their recovery boundaries follow commands, checkpoints, or branches, while failures propagate through dependencies among versioned effects. The trajectory should therefore remain one speculation unit, but recovery should

select a dependency-defined subgraph of tool-step effects. Recording individual steps alone is insufficient because a temporal suffix still follows execution order rather than dependence.

AGENTTX keeps one speculative environment across tool calls while recording their observed effects and dependencies. When a step fails, it selects the affected steps, tracks the object versions needed to remove them, and reconstructs the workspace and agent context from the same survivor set.

We implemented AGENTTX on Linux with dependency capture, object-aware historical reconstruction, fail-closed alias handling, and conversation rewind. Across 144 controlled runs, AGENTTX retained 100% of independent work while removing 100% of invalid descendants. At 64 calls, temporal recovery retained only 41% of independent work. At 48 document entries, preserving valid artifacts avoided 1,425 replay tokens relative to temporal recovery and 3,340 relative to whole-session abort. These results indicate that recovery granularity is not only an execution-overhead problem: reducing isolation cost cannot recover valid work once a coarse rollback boundary has discarded it. We evaluate the prevalence and end-to-end benefit of this opportunity separately from the controlled recovery mechanism.

This paper makes three contributions:

- We identify the recovery-unit mismatch in stateful coding-agent trajectories and characterize the limits of fixed coarse boundaries.
- We build AGENTTX to recover dependency-selected effects while keeping the workspace and agent-visible context aligned.
- We evaluate AGENTTX across recovery correctness, retained work, runtime, replay cost, object identity, and context consistency.

2 Background

2.1 Stateful Coding-Agent Workspaces

Coding agents execute software-engineering tasks as iterative interactions with an operating-system environment. Systems such as SWE-agent and OpenHands let a model inspect repositories, edit files, invoke shell commands, and run tests [2, 3]. A task therefore spans multiple tool calls rather than one isolated program execution. Each call observes repository state left by earlier calls. Shells may also launch compilers, test runners, package managers, and descendant processes whose effects enter the same workspace. The filesystem is therefore persistent execution state: later calls can consume source edits, generated artifacts, build products, and configuration changes.

The broader agent literature makes this execution pattern explicit. Tool-use controllers learn when to invoke external APIs [12]. Reasoning- and-action loops and self-reflection extend trajectories across feedback steps [13]. Interactive benchmarks evaluate multi-turn state and real websites, as in AgentBench [14] and WebArena [15]. Desktop and stateful-tool suites broaden this setting to OSWorld [16] and ToolSandbox [17]; AppWorld evaluates longer application workflows [18], while BrowserGym targets multi-step web interaction [19].

This execution model makes the workspace boundary important. Direct execution exposes every mutation to the host as soon as a tool runs. Recent agent systems instead keep filesystem changes provisional. YoloFS stages session mutations outside the base filesystem and provides snapshots that an agent can return to during self-correction [6]. OpenHands similarly supports sandboxed execution for software-development agents [3]. These systems let an agent continue operating on a mutable workspace while retaining a recovery point.

Repository-level coding systems add another source of long-lived state. They retrieve context across files, localize faults, edit repositories, and execute tests iteratively; RepoCoder [20] and AutoCodeRover [21] illustrate this workflow. Agentless and HyperAgent extend it to autonomous repair [22, 23], while SWE-gym studies longer interactive trajectories [24]. Recent benchmarks broaden the workload from issue fixes to feature development and software evolution [25]. FeatureBench studies feature-level development [26], while SWE-rebench revisits repository evolution [27]. Other suites cover repository-scale work, research replication, and experiment execution; examples include GitTaskBench [28], PaperBench [29], and MLAgentBench [30].

2.2 Speculation and Recovery Units

Existing substrates generally expose recovery at the boundary of the state container they preserve. `try` introduces a semisolate around an opaque component invocation: it captures the command's filesystem effects so they can be inspected, applied, stacked, or discarded before they modify the live system [5]. Sandlock similarly confines a process invocation and uses a copy-on-write work directory to commit or discard its filesystem effects [31]. Agent-native filesystems extend this lifetime across a session. YoloFS retains staged mutations and uses snapshots as restoration points [6], while BranchFS gives each speculative branch an isolated copy-on-write filesystem view with branch-level commit or abort [7].

hS instead targets script performance: it speculatively reorders black-box subprocesses at fine granularity and tracks dependencies to preserve the script's sequential semantics [32]. It is an execution optimization rather than a post-failure recovery mechanism for agent state.

Checkpoint/restore systems preserve still broader execution state. Crab observes turn-level OS effects to decide when a sandbox checkpoint is recovery-relevant, but restoration still returns the sandbox to a selected checkpoint [33]. Waypoint captures filesystem, process, memory, and termi-

nal state, and materializes a new live fork from a selected temporal checkpoint [8]. DeltaBox accelerates checkpoint and rollback of stateful agent sandboxes by storing changes between consecutive checkpoints; its restored state includes both filesystem and process state [9]. Thus, these systems span command effects, staged sessions, branches, and complete session or sandbox checkpoints. Table 1 summarizes the corresponding speculative and recovery boundaries.

Table 1: **Speculation and recovery units.** Existing agent execution substrates preserve and recover state at command, process, session, branch, sandbox, snapshot, or checkpoint granularity.

System	Speculation unit	Recovery unit
`try` [5]	Command	Command
Sandlock [31]	Process	Process
YoloFS [6]	Session	Snapshot
BranchFS [7]	Branch	Branch
Waypoint [8]	Session	Checkpoint
Crab [33]	Sandbox	Checkpoint
DeltaBox [9]	Sandbox	Checkpoint

For the mechanisms in Table 1, the state abstraction determines both speculation lifetime and the native recovery boundary. They span command, session, branch, and checkpoint state, but do not expose a dependency-defined recovery unit within one trajectory.

3 Observation and Motivation

Recovery mechanisms inherit their rollback boundaries from the execution substrate. We first examine what state each boundary can select. A shared multi-turn microbenchmark measures the valid work each boundary sacrifices after a crash, followed by an audit of opportunities in native agent traces.

Artifact availability prevents a complete implementation-to-implementation comparison. We replay the shared cases through pinned artifacts for `try`, BranchFS, YoloFS, and Waypoint. The first two expose native discard or abort operations rather than post-failure agent recovery. Their results therefore measure the state retained by those native boundaries, not the systems' intent. We represent systems without reusable artifacts by their recovery semantics.

Whole branch is the controlled semantic bound for discarding a complete speculative unit. The checkpoint and temporal baselines are defined in the same way, and we assign no implementation latency to these emulations. Named artifact results remain separate from these bounds.

We deliberately favor every coarse baseline with oracle localization. Each receives the exact failed step, and turn recovery receives the exact affected turns. The baseline then selects the smallest set of native units covering all affected effects. This is the maximum retention that its granularity can express. Detection error, imperfect localization, or implementation constraints can only discard additional valid work.

3.1 Analyzing Recovery Paths of Stateful Agents

Execution-defined boundaries. A coding-agent trajectory $T = \langle s_0, s_1, \ldots, s_n \rangle$ comprises ordered tool steps. Each step executes one tool call against state left by preceding steps. Table 1 groups existing recovery paths by the execution boundary they restore. These boundaries encode chronology or control structure; they do not identify which later steps consumed state produced by the failed step.

Consider a failed producer followed by dependent and independent work in different turns. *Root-only* recovery removes the producer but leaves its dependent descendants. A *turn checkpoint* removes complete turns that contain invalid effects. Because their identities are supplied, it selects no additional turns. A *temporal step* removes the global suffix beginning at the producer, and *whole branch* abandons the complete speculative branch. The latter three paths remove the dependent descendants, but they may also remove independent effects enclosed by their execution boundaries.

We include two controls. *No recovery* preserves every effect, including corrupted state. The *ideal oracle* removes only the failed step and the steps that depend on it, preserving all unaffected work. The oracle represents the desired recovered state without prescribing how to discover it.

3.2 Methodology

Controlled cases. We use one filesystem-effect microbenchmark family. A turn contains 16 tool calls, each represented by one recoverable ledger step; a branch contains one to ten turns. The generator varies turn count, fault position, dependency value, graph shape, and interleaving. We analyze 400 paired manifests: 100 vary task length, 90 vary fault position, and 210 vary dependency value. Every policy receives the same manifest and fault root in each paired case.

The generated dependencies identify the failed root and every step that depends on it. Recovery is complete only if it removes all these affected steps. Among complete policies, collateral cost counts removed steps outside this set. We also attribute each step a deterministic token weight to compare the amount of useful work discarded; this weight is not billed model usage or measured replay. We replay all 400 manifests through `try`, BranchFS, YoloFS, and Waypoint to compare recovered states, not cross-environment latency. The first two observations use this shared population.

Native traces. For the third observation, we inspect fault-free SWE-agent runs on SWE-Bench and Terminal-Bench. We include complete traces regardless of verifier outcome; an unsuccessful task attempt still supplies observed tool steps. We treat each workspace-mutating step with a later step as a hypothetical fault root. The analysis connects accesses to the last preceding writer of each included workspace path. Opaque agent-context steps conservatively link earlier observations to later actions. We compare the re-

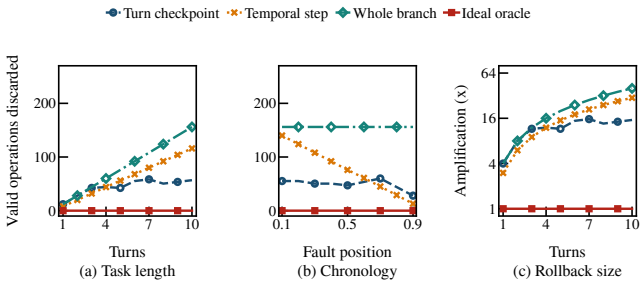


Figure 1: **Correct recovery can still discard avoidable work.** *All plots compare the four correctness-qualified policies. In (a), waste grows with the number of turns for a fixed four-step affected set. In (b), a temporal suffix changes with fault position at fixed task length and affected set. Plot (c) reports rollback amplification. The ideal oracle is an analytic bound, not an implemented recovery policy.*

sulting dependent steps with the temporal suffix that would be removed from the root onward.

We retain tasks and roots with zero observed opportunity in the denominators, and report both per-task incidence and per-root counts. A dependency edge spanning at least five turns is a descriptive slice, not a condition for an opportunity: even a short dependency can leave independent work inside a rollback boundary. This path-level counterfactual excludes some build artifacts and unobserved state. It injects no crash, does not certify that apparently independent steps are independent through agent context, and measures no replay tokens.

3.3 Analyzing Recovery Overheads

Figure 1 exposes the controlled scaling behavior. Its plots retain only policies that remove every affected effect. They hold the ground-truth affected set fixed while varying branch length or fault position, so changes in waste arise from the recovery boundary rather than a larger failure.

With a fixed four-step affected set, increasing the branch from one to ten turns raises discarded valid operations from 12 to 56.8 for turn recovery. Temporal recovery rises from 8 to 116, while branch recovery rises from 12 to 156. The corresponding ten-turn rollback amplification is 15.2×, 30×, and 40×. Moving the fault from 10% to 90% of a fixed ten-turn branch reduces temporal waste from 140 to 13 valid operations. These controlled trends establish that execution structure, rather than affected-set size, drives coarse recovery cost.

Observation #1: Correct coarse recovery still discards independent work. The comparison exposes a correctness–retention tradeoff. Policies that retain nearly all work leave invalid state behind, while correct coarse policies delete valid effects enclosed by their recovery boundaries.

Figure 2 quantifies this tradeoff over all 400 paired manifests. No recovery removes 0% of the affected steps, while root-only undo removes 17.03% on average. Their low rollback cost therefore reflects under-recovery, not efficiency.

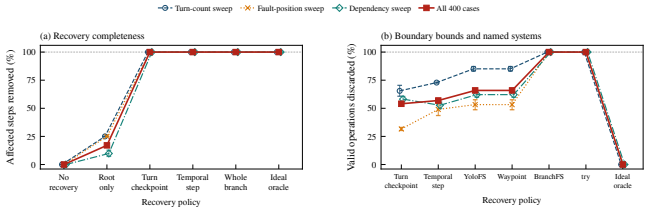


Figure 2: **Recovery tradeoff across 400 paired cases.** *Lines report the turn-count (100 cases), fault-position (90 cases), dependency-value (210 cases), and combined-suite means; error bars give 95% bootstrap confidence intervals. The left plot applies the correctness gate to boundary policies. The right plot compares optimistic boundary upper bounds with artifact-native* `try`*, BranchFS, YoloFS, and Waypoint, each labeled by system name.*

Turn checkpoint, temporal step, whole branch, and ideal recovery remove 100% of the affected set in every case.

Among the semantic policies, checkpoint, temporal, and branch recovery discard 54.02%, 56.90%, and 100% of valid work on average. The ideal boundary discards none. Artifact-native YoloFS and Waypoint each discard 65.89%, 11.87 percentage points more than the oracle-aided turn boundary. BranchFS and `try` discard the complete speculative unit through their native operations. The recovered states agree within each pair across all 400 manifests. Correctness therefore does not imply efficient recovery: coarse boundaries restore safety by deleting independent progress. We first test completeness, then compare the valid work preserved by complete policies.

The turn policy already uses a favorable boundary: it rolls back only turns that contain affected steps. Its 54.02% collateral loss therefore comes from valid steps inside those turns. Substantial avoidable waste remains within the turn, below the finest boundary exposed by turn-level recovery.

Observation #2: Temporal recovery wastes work because time and dependency order diverge. The desired recovered state is usually not a temporal prefix.

Figure 3 exposes this mismatch directly. Coarse recovery follows execution order, whereas failure propagation follows dependencies. When valid and affected effects are interleaved, a temporal boundary cannot separate them. Across the 400 paired manifests, only 10 exact survivor states are temporal prefixes. In the other 390 (97.5%), no single checkpoint removes every affected effect while retaining every valid effect.

To vary the severity of this mismatch, let r be the failed step and S the steps after it. Dependency value is the fraction of S that depends on r. A value of zero means no later step depends on the failure; one means every later step does. The figure sweeps this value from 0 to 1 in a five-turn, 80-operation branch. The exact target is non-prefix in 200 of these 210 cases; only the ten fully dependent cases form a suffix.

At dependency value 0, checkpoint, temporal, and whole-branch recovery discard 660, 2,596, and 3,476 units of useful token weight. They retain 81.01%, 25.32%, and 0% of

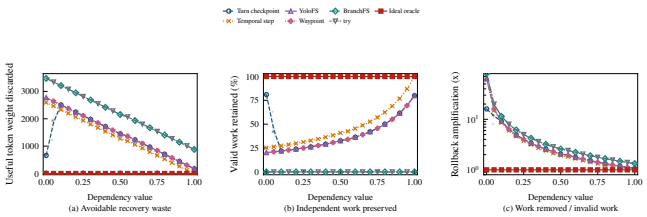


Figure 3: **Recovery waste reflects temporal–dependency mismatch.** *Across all 21 dependency values and ten paired manifests per point, sparse dependencies create the largest gap between execution order and failure dependence.* `try`*, BranchFS, YoloFS, and Waypoint are artifact-native and labeled directly. YoloFS overlaps Waypoint, while BranchFS overlaps* `try`*; distinct markers identify both measurements. Other lines are optimistic boundary semantics. Lines show means, faint points show paired cases, and shaded bands give 95% paired-bootstrap confidence intervals. Token weight is deterministic work attribution, not billed model usage.*

valid work, with rollback amplification of 16×, 60×, and 80×. Artifact-native YoloFS and Waypoint recover identical states in all 210 dependency cases. At dependency value 0, each discards 2,772 units, retains 20.25%, and incurs 64× amplification. BranchFS and `try` match whole-branch recovery because their native operations discard the complete unit. At dependency value 1, YoloFS and Waypoint each discard 176 units and retain 80%. Temporal recovery becomes exact, checkpoint discards 176 units, and BranchFS and `try` each discard 880. The ideal oracle remains at zero useful work discarded, 100% retention, and 1× amplification across the sweep. Denser checkpoints or faster branch aborts can reduce mechanism overhead, but they cannot represent a non-prefix survivor state. Eliminating this waste requires selecting recovery state by dependencies rather than by execution time.

This result locates a granularity gap. Mechanisms that preserve a long-running agent workspace recover at a turn, checkpoint, branch, or session boundary. Command-scoped semisolates expose one call's effects but end the speculative unit at that call. A long-lived runtime must instead retain the effects of individual tool steps across the trajectory. Step-level recording is necessary, but the next observation shows that it is not sufficient.

Observation #3: Long-range dependence alone does not imply a large recovery opportunity. We next inspect fault-free agent trajectories to test whether the structural opportunity constructed above appears in real executions. This analysis treats each observed workspace write as a hypothetical fault root; it does not inject a crash or measure replay tokens. We use SWE-agent because the current Codex CLI exporter groups its entire execution into one turn, preventing a cross-turn comparison.

Figure 4 separates long-distance dependencies from independent work that a temporal suffix would discard. Cross-turn workspace dependencies occur in 171 of 189 complete SWE-Bench traces and 44 of 54 Terminal-Bench traces. Edges spanning at least five turns appear in 99 and 15 traces.

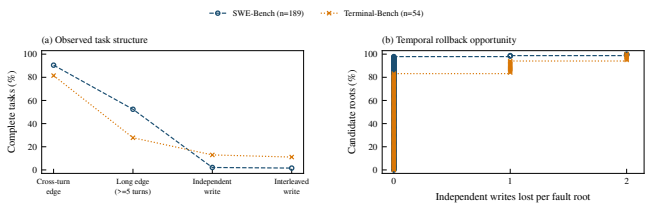


Figure 4: **Observed recovery opportunities in native SWE-agent traces.** *This interim full-sweep snapshot contains 189 complete SWE-Bench and 54 complete Terminal-Bench traces, including failed task attempts. A long edge spans at least five turns; an interleaved write lies outside the observed dependency closure but between affected steps. Panel (b) considers 236 and 101 hypothetical write-root candidates, respectively. These are observed-path counterfactuals, not certified independent work or measured replay cost.*

Yet only 4 and 7 traces, respectively, contain an observed later independent write; just 3 and 6 place such a write inside the affected execution interval. Correspondingly, 231 of 236 SWE-Bench candidate fault roots and 84 of 101 Terminal-Bench roots have no additional independent write for temporal recovery to discard.

Thus real trajectories exhibit the temporal–dependency mismatch, but a long dependency span is not itself evidence of large avoidable loss. The observed opportunity is sparse and workload-dependent. Our path analysis excludes some build artifacts and cannot certify independence through agent context or unobserved process state; the figure bounds a possible opportunity rather than demonstrating realized savings. The controlled cases above isolate the mechanism, while real-agent recovery must establish its end-to-end benefit separately.

Significance. Together, the observations progressively narrow the required recovery model. Observation #1 excludes policies that appear efficient by leaving invalid state behind and exposes substantial collateral loss within affected turns. Observation #2 attributes that loss to disagreement between temporal order and dependencies. Observation #3 shows that long-range dependencies occur in native trajectories, while the independently writable work that could be retained varies sharply across tasks. The required recovery unit is therefore a dependency-selected set of tool-step effects, not another fixed interval in execution time.

3.4 Motivation

Attempts. The three observations rule out refining a temporal boundary as a complete solution. No recovery and root-only undo fail the correctness gate. Turn checkpoints and branch abort remove invalid state but discard valid work inside their boundaries. Temporal-step recovery uses a finer unit, yet still removes a suffix. The native-trace audit shows that the possible benefit depends on whether independent work actually overlaps the affected execution interval.

Insight. The observations show that a turn is too large to serve as the recovery unit. Each tool step and its OS ef-

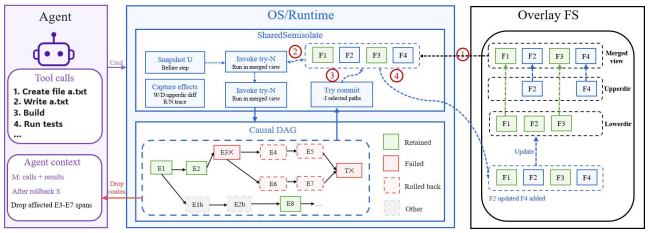


Figure 5: **Execution and selective recovery in AGENTTX.** *The agent executes tool calls against a shared overlay view. The runtime records their effects and dependencies, removes the failed step and its descendants from the workspace and agent context, and retains independent work.*

fects should instead be recorded separately. Step granularity alone is insufficient: recovery must select these records by dependency rather than remove a temporal suffix. The trajectory remains the speculation unit, while a dependency-defined subgraph of step effects becomes the recovery unit. This separation preserves independent effects on both sides of a failed step.

Design challenges. Realizing selective recovery proposes three challenges: 1) *dependency discovery*, uncovering relations hidden inside shells, compilers, tests, and descendant processes; 2) *object identity*, tracking objects through aliases, renames, replacements, and layered filesystems; and 3) *selective reconstruction*, rebuilding a state that may never have existed at a checkpoint and aligning the agent-visible context with it. The next section states the recovery model used to describe these requirements.

4 Selective Recovery Model

To describe the separation observed in Section 3, we model the trajectory as an *Agent Effect Transaction* (AET). This notation gives execution and recovery different units: the trajectory remains speculative, while recovery selects a dependency-defined subset of versioned effects. The same selection applies to the workspace and agent-visible context. Section 5 maps this model to Linux mechanisms in AGENTTX.

4.1 Overview and Design Principles

Execution and effect capture. Figure 5 connects agent tool calls and context (left), the runtime (center), and the overlay filesystem (right). Before a call, AGENTTX snapshots the upper layer, then runs the command through `try` in the merged view. In the example, the merged view takes F1 and F3 from the lower layer while the upper layer updates F2 and adds F4. After the call, an upper-layer diff records writes and deletes; tracing records reads and failed lookups.

Dependency-guided recovery. The captured effects form a DAG of tool steps. In the figure, failed E3 invalidates E4–E7, but later E8 remains on an independent path. AGENTTX rebuilds the overlay view without the invalid effects and removes their associated calls and results from the agent context. E8 and its context remain available. The separate `try` commit path publishes selected surviving files to the host; it does not determine the rollback set.

Design principles. An AET spans the complete tool-call trajectory rather than one command or recovery attempt. It records dependencies from observed reads, absence, writes, and deletes rather than inferring them from command text. Recovery reconstructs object versions without the selected steps while preserving supported retained effects. The model-visible history follows the same survivor set.

4.2 Recovery State and Transitions

Scope. We model workspace-local filesystem effects and the agent-visible tool-call history produced by trusted but fallible coding agents. Network services, hidden provider state, process memory, terminal state, and adversarial confinement remain outside the transaction.

State. An AET is the tuple $\langle V, L, H, M \rangle$. V is the current speculative filesystem view. L records tool calls, typed effects, dependencies, and status. H stores historical object states needed for reconstruction. M records agent-visible message spans and their associations with ledger steps.

The tuple assigns one role to each recovery concern. V provides cross-call workspace state. L determines causal recovery sets. H supplies versions that can remove selected effects. M determines which tool results remain valid after rollback.

Transitions. *Append* executes one tool call in V, appends its effects to L, and binds the corresponding request and result in M. *Recover* computes a causal closure, reconstructs the affected portion of V from H, and rewinds M to the matching survivor set.

Invariants. Every successful recovery preserves two properties. *Selective recovery* removes every selected effect while preserving supported, non-overlapping retained effects. *Workspace–context agreement* removes invalid tool results from M while retaining messages associated with surviving effects.

These semantics do not require a particular tracer, snapshot representation, or provider message API. The remaining subsections define the recovery set and the reconstruction of workspace and agent-visible context.

4.3 Causal Recovery Semantics

Effect records. Each ledger node $s_i = (i, \text{tool}_i, E_i, P_i, \text{status}_i)$ represents one tool step. E_i contains its observed filesystem effects and P_i contains the identifiers of earlier steps on which it depends through either workspace state or an agent-visible result. Effects include successful reads (R), negative lookups (N), writes (W), and deletes (D).

Dependency rule. Let $\text{overlap}(p, q)$ hold when two effect names identify overlapping filesystem objects. For an active earlier step $j < i$, the transaction adds $j \in P_i$ when:

- s_j writes or deletes p, and s_i reads, negatively looks up, writes, or deletes an overlapping q; or

- s_j negatively looks up p, and s_i later writes an overlapping q; or
- the request for s_i consumes a value returned to the agent by s_j.

The write/delete rule captures dependencies induced by observed or subsequently modified state, including read-after-write and write serialization. The negative-lookup rule captures dependencies on absence: a later creation can invalidate namespace state observed by an earlier negative lookup. The final rule captures values that flow through the agent context without appearing in a filesystem read.

Object relation. Path equality alone does not define overlap. We make object identity explicit with an abstract relation $\equiv_I$ over effect names. Path equality implies overlap, while aliases may also overlap when the identity relation can establish that they refer to the same object.

A descriptor read is associated with the object opened earlier only when observation supplies that relation. Otherwise, the effect remains unknown instead of receiving a guessed dependency. Selective recovery that requires an unresolved identity relation must fail closed and lies outside the recovery guarantee. The identity relation is part of the recovery model; Section 5 describes how AGENTTX approximates it on Linux.

Recovery set. For a failed active step f, the transitive descendant closure is

$$\begin{aligned} C_0(f) &= \{f\}, \\ C_{k+1}(f) &= C_k(f) \cup \{s_i \mid P_i \cap C_k(f) \neq \emptyset\}, \\ C(f) &= \bigcup_{k \geq 0} C_k(f). \end{aligned} \tag{1}$$

$C(f)$ is the least transitive descendant closure of f among active steps. It may contain steps that are separated in time by independent work. Temporal rollback instead selects the contiguous suffix $\{s_i \mid i \geq f\}$, regardless of whether each later step depends on f.

4.4 Object-Aware Reconstruction

Target version. Computing $C(f)$ identifies which effects should no longer contribute to the speculative state, but removing their latest writes is not sufficient when the same object has been modified repeatedly. Recovery must instead reconstruct the version that would remain after excluding the selected recovery subgraph.

Let $W(C)$ denote the objects written or deleted by steps in $C(f)$. For each target in $W(C)$, recovery obtains from H a historical state that predates its earliest selected effect. The retained active history is then considered against the selected targets so that later independent state is not discarded solely because it occurs after the failed step.

Retained overlap. A retained effect can nevertheless conflict with selective reconstruction. If a retained active effect overlaps a target in $W(C)$ and the model cannot establish a supported reconstruction, recovery rejects the operation rather than guessing a merge. This fail-closed rule trades availability for the following safety invariant:

Recovery guarantee. After successful recovery within the stated identity and reconstruction contract, no write or deletion from $C(f)$ remains visible. Every supported, non-overlapping retained effect remains visible in the speculative view.

Only after reconstruction succeeds are the selected ledger nodes marked as rolled back. Reconstruction therefore changes the workspace by causal membership rather than restoring a complete temporal snapshot. Unreachable historical states may then be reclaimed.

4.5 Agent-Context Reconstruction

Message association. M groups each tool request and result into a span associated with its ledger step. If a later request copies a value from an earlier result, the runtime adds that relation to P_i, even when the value never enters the filesystem.

Shared recovery set. The closure $C(f)$ follows both filesystem and message dependencies to a fixed point. The runtime removes the filesystem effects and message spans of exactly those steps, plus stale recovery-control messages, while retaining independent steps on both sides of the failure.

Reconstructed view. The next model request is assembled only from active spans and an authoritative notice that identifies retained and invalidated effects. The reconstructed workspace and model-visible history therefore describe the same survivor set. Hidden provider state is outside this guarantee; black-box agents receive a retained-effects manifest instead of an in-place message rewrite.

5 AgentTX Implementation

AgentTX maps each model component to a Linux mechanism. A shared OverlayFS view keeps the trajectory in V; effect capture constructs L; the version store and object catalog preserve H; a ledger-bound conversation log maintains M. Together, these mechanisms preserve trajectory-wide speculation while restricting rollback to a dependency-defined effect subgraph. We follow this path from execution through fault localization to workspace and context recovery, then describe performance optimizations.

5.1 Runtime Architecture and Continuous Speculation

Workspace construction. For each tool call T_i, AgentTX selects the project baseline as the read-only `lowerdir`, creates the session's writable `upperdir`, and allocates the OverlayFS `workdir` and `merged` mount point. The command runs against `merged`, not the host project.

OverlayFS resolves reads against the upper layer and then the baseline. A write to a lower object triggers copy-up, a new object enters the upper layer, and deletion creates a whiteout. The kernel uses `workdir` for internal bookkeeping. AgentTX configures these directories and controls their lifetime; OverlayFS performs lookup, copy-up, and whiteout

operations.

Cross-call state. All calls in a session reuse one `try -N` upper layer. Later calls therefore observe earlier provisional effects, while the base workspace remains unchanged. The launcher synchronizes `$PWD` with the protected workspace so the process working directory and environment refer to the same view.

The runtime now has one call's effects inside the shared trajectory, but not their cross-call dependencies. Ledger construction adds the edges that define the recoverable subgraph.

5.2 Effect Capture and Ledger Construction

Mutation effects. AgentTX fingerprints the unmounted upper layer before and after each command. Content and metadata fingerprints capture writes, chmod, chown, touch, empty directories, symlinks, and deletions. OverlayFS character-device and `.wh.*` whiteouts are both normalized to *D* effects.

Observation effects. AgentTX uses `strace` [34] to capture successful reads and ENOENT/ENOTDIR lookups. The parser follows subprocess creation and working directory changes. A failed trace or parse aborts the call unless observation is explicitly disabled. Trusted tools may instead declare effects, making the declaration part of the trusted computing base.

Path resolution and coverage. The parser resolves supported pathname operations relative to the current directory or a known file-descriptor origin. It retains both requested and resolved names when a symlink changes the path. A non-`AT_FDCWD` operation without a known descriptor origin fails closed. Modeled Linux pathname syscalls define the coverage boundary; the trace does not prove that every possible filesystem access was observed.

Ledger construction. At command return, AgentTX merges the observed *R* and *N* effects with the derived *W* and *D* effects. It normalizes paths and supported identities, preserves observable order within the call, and appends one record E_i. The ledger then compares E_i with active earlier records using the rules in Section 4.3. Unsupported identities remain unknown rather than receiving dependencies inferred from command text.

5.3 Historical Versions and Object Identity

Version store. Before step *i*, the layer store records `before_i`; after the call, it records the resulting upper-layer state as V_i. Regular-file contents are stored as immutable content-addressed blobs and linked into snapshots. Native whiteouts, directory metadata, and mode-000 trees are preserved. These states provide the pre-effect versions required by Section 4.4.

Namespace normalization. AgentTX treats ancestor-related paths as overlapping and retains lexical and resolved names for supported symlink accesses. A rename contributes a delete at the old name and a create at the new name. These rules cover namespace relations that can be established from the observed operation.

Hard-link identity. The active OverlayFS path enables `index=on` to preserve hard-link visibility during live copy-up. A persisted object catalog connects accesses through known sibling names and records tested upper-created link groups. Recovery expands a known alias group before checking overlap or reconstructing an object.

Identity boundary. AgentTX promotes a hard-link relation only when the substrate or upper-layer materializer verifies it. Bind mounts, external aliases, arbitrary generation changes, and general multi-object identity remain outside the contract. An operation that requires one of these relations fails closed.

5.4 Fault Localization and Recovery Planning

Failure observation. A failed verifier establishes that the current view is invalid, but does not identify the faulty step. AgentTX records the failed check as a ledger step and uses its observed dependencies to find earlier active steps that changed state. The search excludes committed steps and, when available, steps before the last passing check. Dependency distance and diagnostic references rank candidates; neither constitutes proof that a candidate is the fault's origin.

Counterfactual checks. For each candidate, AgentTX forks the speculative view, removes the candidate and its dependents, and reruns the same verifier in the fork. It can also test candidate groups to handle interacting faults. Among plans that pass within the probe budget, it selects one that discards the least active work, with conversation loss as a secondary cost. A passing check establishes a sufficient recovery plan for that verifier, not a unique root cause. If no plan passes, localization is inconclusive and the causal path does not change the live view. The real-workload causal evaluation disables optional coarse fallback so an inconclusive search cannot count as causal recovery.

5.5 Causal Reconstruction

Recovery plan. AgentTX applies a verified plan to the live view only after counterfactual checks finish. For its selected root steps, it alternates ledger and conversation expansion until the recovery set reaches a fixed point. It collects objects written or deleted by that unified closure, expands known alias groups, and chooses the version preceding each object's earliest selected effect.

Retained-state check. The runtime compares every target with active effects outside the closure. A non-overlapping retained effect can survive. An overlap without a supported object-level reconstruction rejects recovery before the speculative view is changed.

Reconstruction. AgentTX builds a fresh upper generation from the selected historical versions and rebases retained state outside the closure onto it. It validates every affected target before swapping the generation into the live session. Ledger nodes become rolled back only after the

swap succeeds. Recovery reconstructs the view from effect history; it does not replay the trajectory.

5.6 Conversation Reconstruction

Ledger-bound spans. AgentTX persists each native-agent tool request and result as a call span bound to its ledger step. The conversation log also records message dependencies when a later tool request copies a distinctive value from an earlier result.

Synchronized rewind. Workspace reconstruction and conversation rewind consume the same fixed-point recovery set. After reconstruction succeeds, the runtime invalidates the corresponding spans and stale control messages, then rebuilds the next provider request from active spans.

Recovery handoff. Native tool-calling agents resume from the reconstructed message list plus a ledger-derived notice naming retained and invalidated state. External black-box agents cannot expose mutable provider history, so they resume from a compact retained-effects manifest. AgentTX does not claim to restore hidden reasoning state or process memory.

5.7 Performance Engineering

Persistent execution. A Python worker exchanges length-prefixed requests inside one semisolate. This amortizes namespace and overlay setup across a long trajectory. If the worker dies, the current request uses the one-shot path and the next request starts a new worker. Worker reuse does not change recovery semantics.

Incremental history. After the initial snapshot, AgentTX clones the previous snapshot and replays only paths changed by the latest call. Content-addressed blobs deduplicate regular-file contents across versions. These optimizations reduce history maintenance while preserving the same H states used by reconstruction.

6 Evaluation

Our evaluation answers four questions:

RQ1 Does causal recovery remove invalid descendants while retaining independent work (Section 6.2)?

RQ2 What runtime cost do trajectory isolation and dependency capture add (Section 6.3)?

RQ3 Which capture, identity, workspace-reconstruction, and context-rewind mechanisms are necessary (Section 6.4)?

RQ4 Can a live agent use causal recovery, and how many replay tokens does retained work avoid (Section 6.5)?

6.1 Setup and Methodology

The experiments run on an $x86_{64}$ server with an AMD EPYC 7713. The server has 32 online CPUs, 62 GiB of memory, and Linux 5.4.0-216-generic. The prototype uses upstream `try`, OverlayFS, `strace`, and Python 3.8.10. Unless stated otherwise, live-agent and controlled-replay experiments in this section use `deepseek-v4-flash`; the 400-manifest model-token study uses GPT-5.6 Luna. Official SWE-Bench and Terminal-Bench recovery runs use the same AgentTX overlay; the reported application cells apply the official gold patch or Terminal-Bench oracle after the selected policy.

Application evaluation uses two official suites and no other application workloads. SWE-Bench Lite supplies three issue-resolution instances (`pytest-dev__pytest-8906`, `pallets__flask-4992`, `pylint-dev__pylint-5859`) [4]. Terminal-Bench supplies three shell tasks (`hello-world`, `csv-to-parquet`, `log-summary`) [35]. Each run starts from the official snapshot and overlays a recovery DAG containing independent notes, a faulty producer, and a derived artifact. It then applies causal, temporal, or whole-session recovery. The validator combines the official verifier with independent-document retention.

The remaining measurements are mechanism microbenchmarks, not additional application workloads.

Every run starts from a fresh workspace. Correctness validators inspect the speculative view and agent-visible context before and after recovery. The fixed comparison uses 50 independent samples per mode; the causal and token sweeps use three repeats per aggregated configuration. The multi-step runtime comparison uses two repeats per mode and length at 32/64/96/128 calls on the isolation microbenchmark, matching the motivation sweep. The live recovery experiment uses three fresh sessions.

The official application suite uses one oracle-repair run per task and policy. The coverage matrix uses three repeats per topology cell.

We call a run *causal-correct* when recovery removes the invalid producer and every derived artifact, preserves independent work, and leaves the agent-visible context consistent with the workspace. Modes without causal recovery can still measure execution cost, but they do not satisfy this joint predicate.

We separate baselines by the property they measure. Bare execution is an unsafe latency lower bound because it writes to the host immediately. Per-call `try`, session `try`, shared `try`, shared checkpoints, and Bubblewrap isolate different execution costs but do not implement causal recovery. The no-trace configuration removes read dependencies and serves as a correctness ablation. Temporal checkpoint and whole-session abort emulate coarser recovery over the same trajectory; they are not external-system runs.

6.2 RQ1: Causal Recovery Semantics

We hypothesize that a dependency-defined recovery unit preserves independent work that a temporal unit discards. Both units should remove the failed producer and its descendants. Without dependency capture, the ledger should instead preserve derived corruption as apparently independent work.

The sweep executes filesystem effects through the real overlay rather than simulating graph nodes. Its 48 aggre-

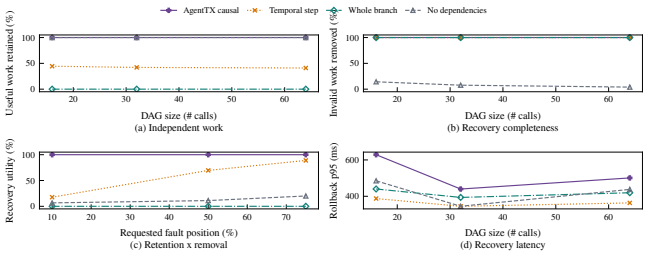


Figure 6: **Dependency-defined recovery separates correctness from collateral loss.** *Across 144 controlled runs, full AgentTX retains 100% of independent work and removes 100% of invalid work. At 64 calls, temporal recovery retains 41%, while the no-dependency ablation removes only 4% of invalid descendants.*

gated configurations span lengths of 16, 32, and 64; chain, fan-out, and layered DAGs; three fault positions; and three independent-work ratios. Causal, temporal, and whole-session recovery receive identical declared reads, while the fourth mode removes read dependencies. Three repeats produce 144 runs, each validated for host cleanliness, independent retention, and invalid-descendant removal.

Figure 6 reports both retention and invalid removal for our system against the same baselines. Full causal recovery reaches 100% on both metrics in every tested configuration. At 64 calls, temporal recovery removes the invalid work but retains only 41% of independent work; whole-session abort retains none. The no-dependency ablation retains independent files but removes only 4% of invalid descendants.

Causal rollback has a p95 latency of 238.9 ms at 64 calls. The separate ten-write predicate also succeeds in 50 of 50 full-system runs and zero of 50 runs for every other measured mode. These experiments validate the recovery-unit claim on controlled effect DAGs, not arbitrary repository failures or non-filesystem effects.

We also examine model-token cost on the same 400 multi-turn manifests, with three repetitions per manifest and policy. This separate run uses GPT-5.6 Luna at maximum reasoning effort. All seven policies have measurements for their 1,200 cells. Figure 7 includes every recorded attempt, including failed runs; a failed run with no reported model usage contributes zero rather than an imputed cost. Completion requires both the final-state verifier and the replay-operation protocol.

Across all cells, clean execution averages 123K tokens, the ideal oracle 606K, and AgentTX 676K. The checkpoint, temporal-step, whole-branch, and no-rollback controls average 1.14M, 1.62M, 2.36M, and 2.71M tokens. These are charged means, not cost per successful recovery. AgentTX completes 1,193 of 1,200 cells, compared with 1,191 for ideal, 1,161 for checkpoint, 705 for temporal, 441 for whole branch, and 1,162 for no rollback. In particular, temporal and whole-branch runs often reach a correct final state but fail the operation protocol; treating their lower-usage failures as efficient recoveries would confound cost with completion. On cells where both policies complete, AgentTX uses 531K

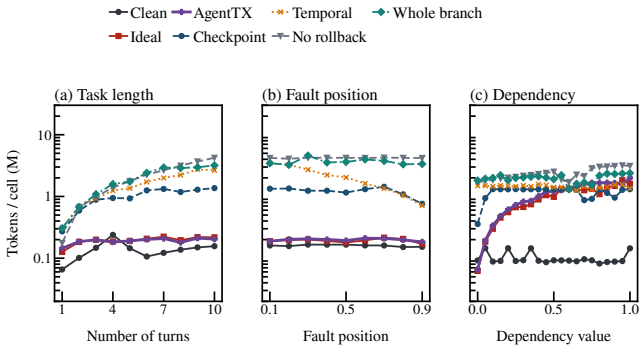


Figure 7: **Measured model-token cost on the common microbenchmark.** *Each point averages 30 case–repetition cells, including failures, across the same manifests. Zero-usage failures remain zero in the charged-token mean; the curves must be read with completion rates in the text. The ideal oracle receives the fault location without running a localizer.*

fewer tokens than checkpoint on average (1,154 pairs) and 1.01M fewer than temporal step (699 pairs). These paired comparisons avoid mixing different completed-case populations, but do not price an unsuccessful run's eventual retry.

6.3 RQ2: Runtime Cost and Scaling

We hypothesize that one persistent speculation unit amortizes setup across the trajectory. Dependency capture should add cost because it observes reads and retains a larger effect graph. Unsafe bare execution remains a latency lower bound, not a correctness-equivalent baseline.

Table 2 reports the 64-call slice of the same comparison used in the motivation section (Section 3), now including the AgentTX series on the identical isolation microbenchmark. Full AGENTTX costs 90.9 ms/step, 94.6% less than per-call `try` at 1684.9 ms/step and 94.5% less than the snapshot and whole-branch proxies at 1659.7/1644.9 ms/step. It remains 10.8× slower than bare execution at 8.4 ms/step, which pollutes the host immediately. The 18.1 ms/step difference from the no-trace ablation (72.8) includes both system-call tracing and the additional dependency state. Relative to the Crab- and DeltaBox-style checkpoint floors (74.6 and 75.5), full AGENTTX adds 15.4–16.4 ms/step while additionally providing the causal recovery semantics that those floors do not implement.

We separately measure runtime distributions on a fixed ten-write trajectory. Each mode runs 50 times on fresh workspaces with `strace` selected for the full configuration. Table 3 reports the distribution, and Figure 8 pairs p99 latency with the causal predicate. These values are not compared numerically with the mixed-workload comparison above.

Table 4 reports the same 32/64/96/128-call sweep for every mode. Full AGENTTX falls from 133.5 to 65.1 ms/step as it amortizes fixed setup. The Crab-style and DeltaBox-style floors fall from 127.8 to 47.3 and from 129.6 to 47.6 ms/step, respectively. Total AGENTTX wall time still

Table 2: **Runtime comparison at 64 calls.** *Every mode of the motivation sweep plus the AgentTX series on the same isolation microbenchmark. Boundary names denote local runtime emulations, not artifact-native measurements. Full AgentTX is 94.6% below per-call* `try` *and 10.8× above unsafe bare execution.*

Mode	ms/step	Host clean
Bare	8.4	no
Per-call `try`	1684.9	yes
Snapshot proxy	1659.7	yes
Whole-branch proxy	1644.9	yes
Crab-style	74.6	yes
DeltaBox-style	75.5	yes
AGENTTX no-trace	72.8	yes
AGENTTX full	90.9	yes

Table 3: **Runtime distribution on the fixed ten-write trajectory.** *Full AgentTX reaches 68.169 ms/step at p99, 66.4% below per-call* `try` *at 202.746 ms/step, and satisfies the causal predicate in all 50 runs.*

Mode	Mean	p50	p95	p99	Correct
		ms/step			runs/50
Bare	2.164	2.095	2.323	4.976	0/50
Per-call `try`	193.726	193.826	201.427	202.746	0/50
Session `try`	18.621	18.232	21.988	22.390	0/50
Shared `try`	184.230	184.204	190.712	196.776	0/50
Shared checkpoint	47.186	46.599	51.206	52.147	0/50
Bubblewrap	1.145	1.162	1.274	1.301	0/50
AGENTTX no-trace	50.395	49.818	54.100	55.508	0/50
AGENTTX full	58.757	57.738	63.511	68.169	50/50

rises from 4.3 to 8.3 s, so a lower per-step cost does not make a longer run faster.

Per-call `try` and the two overlay proxies remain at 1631.1–1684.9 ms/step. Their setup is repeated rather than amortized. The no-trace ablation falls from 122.4 to 48.7 ms/step. Full AGENTTX remains 11.1–18.1 ms/step above it, exposing the cost of dependency capture.

The repeated ten-write experiment establishes a stable distribution only for that fixed trajectory. The ten-write and multi-step scaling results use different command mixes, so their absolute values are not interchangeable. Across both studies, persistent execution lowers repeated isolation cost, but tracing and heavy tool boundaries still dominate the tail.

6.4 RQ3: Recovery Mechanism Evidence

We hypothesize that dependency-defined rollback requires typed effect capture, object-aware history, workspace reconstruction, and context rewind. Removing one mechanism should retain invalid descendants, lose an alias relation, restore the wrong object version, or leave stale tool results in the model context.

The 64-call dependency validator injects fault roots at steps $[27, 29, 30]$. Full AGENTTX identifies all three, removes the faulty formatter and derived report, and retains the independent documentation, configuration, and repair. Without read tracing, the ledger identifies only step 27 and leaves the report. This connects the 4% ablation result in RQ1 to a concrete missed edge.

A separate 20-step no-op microbenchmark isolates capture cost. Enabling tracing raises mean latency from 295.63

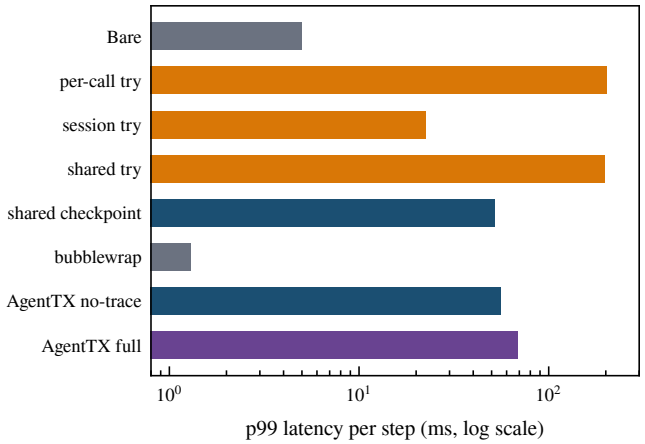


Figure 8: **Persistent execution reduces the p99 isolation cost.** *Across 50 ten-write runs, full AgentTX reaches 68.169 ms/step, 66.4% below per-call* `try`*, and is causal-correct in every run.*

Table 4: **Multi-step runtime sweep (ms/step).** *Every mode at 32/64/96/128 calls on the same isolation microbenchmark, two repeats each. External-system names denote local boundary emulations. Full AgentTX amortizes setup like the checkpoint floors while per-call* `try` *and the overlay proxies stay flat.*

Mode	32	64	96	128
Bare	8.6	8.4	8.1	8.0
Per-call `try`	1659.9	1684.9	1670.8	1664.2
Snapshot proxy	1636.4	1659.7	1638.8	1631.1
Whole-branch proxy	1663.4	1644.9	1649.6	1664.3
Crab-style	127.8	74.6	56.5	47.3
DeltaBox-style	129.6	75.5	56.8	47.6
AGENTTX no-trace	122.4	72.8	56.1	48.7
AGENTTX full	133.5	90.9	72.9	65.1

to 319.43 ms/step across three repeats (+23.80 ms, +8.0%). Shell and test commands access more paths, so this is not a workload-wide tracing tax.

Table 5 states the necessity claims. Table 6 reports the corresponding topology matrix with three repeats per cell. Supported cases include ordinary paths, `openat` with `AT_FDCWD`, negative lookups, symlink aliases, rename, pre-existing hard links, and upper-created hard links. All supported cases pass. External aliases fail closed.

Bind mounts are unavailable without privilege on this host and are recorded as such rather than as silent success. Fd-relative opens through a non-`AT_FDCWD` directory descriptor either resolve inside the workspace or omit a cwd-invented dependency.

The 33-test recovery-state and conversation suite passes in full. It applies causal, temporal, and whole-session rollback to matched workspace and message histories, verifying that each policy removes the corresponding tool-result spans and preserves eligible spans. It also covers dependency chains that move from a tool result, through the agent context, into a later filesystem effect. The unified closure removes those effects and gives the next model request a history consistent with the reconstructed workspace.

Table 5: **Which mechanisms are necessary.** *Each row names the failure mode if the mechanism is removed and the primary evidence used in this section.*

Mechanism	If removed	Evidence
READ / NEGATIVE capture	derived corruption retained	RQ1 ablation; 64-call validator
Object identity	sibling alias missed during rollback	coverage matrix
Historical reconstruction	wrong version after partial rollback	same-path recovery tests
Context rewind	stale tool result remains visible	conversation consistency tests

Table 6: **Capture and identity coverage.** *Supported cells require correctness; unsupported cells require fail-closed behavior. Three repeats per case.*

Case	Contract	Pass rate
Ordinary path	correct	3/3
`openat(AT_FDCWD)`	correct	3/3
Negative lookup	correct	3/3
Symlink alias	correct	3/3
Rename	correct	3/3
Pre-existing hard link	correct	3/3
Upper-created hard link	correct	3/3
External alias	fail-closed	3/3
Bind mount	unavailable / fail-closed	n/a
Fd-relative dirfd	resolve or omit	3/3

6.5 RQ4: Live-Agent Recovery and Avoided Replay

We hypothesize that a tool-selecting model can use the dependency ledger without a scripted rollback target. We also expect retained independent artifacts to reduce model work after recovery. We separate three measurements: control-plane use, avoided replay tokens under a fixed regeneration protocol, and official SWE-Bench / Terminal-Bench recovery.

Control plane. The live benchmark seeds one repository with a faulty pipeline, an independent release note, a derived invalid artifact, and failing tests. The model must inspect the ledger, choose the earliest faulty producer, invoke causal rollback, retain the note, remove the artifact, repair the task, and pass tests. The validator then checks workspace–context agreement.

Figure 9 summarizes three fresh sessions. Root selection, rollback target, independent retention, invalid removal, and tests succeed in all three. The recovered context contains no invalidated tool result, and wall-time p50/p95 is 42.9/44.4 s with `deepseek-v4-flash`. This covers one seeded task and one model.

Avoided replay tokens. The controlled token study uses a five-step trajectory with two valid documents interleaved with one faulty producer and its descendants. Causal recovery retains both documents, an optimistic temporal checkpoint loses one, and whole-session abort loses both. A real model regenerates only the missing documents at 12, 24, and 48 entries, with three repeats per policy and size.

Figure 10 reports all 27 samples. Every sample succeeds without a host leak. Causal recovery needs no regeneration call and avoids between 864 and 1,425 tokens relative to temporal recovery, and between 1,797 and 3,340 relative to whole-session abort. These are avoided replay tokens, not total session-token savings, and the coarse policies are recovery-granularity emulations rather than external artifacts.

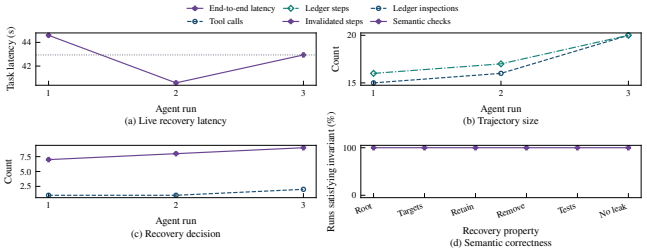


Figure 9: **A live agent uses causal recovery.** *In all three sessions, the agent selects the faulty root, retains the independent artifact, removes invalid descendants, passes tests, and resumes from a context consistent with the workspace.*

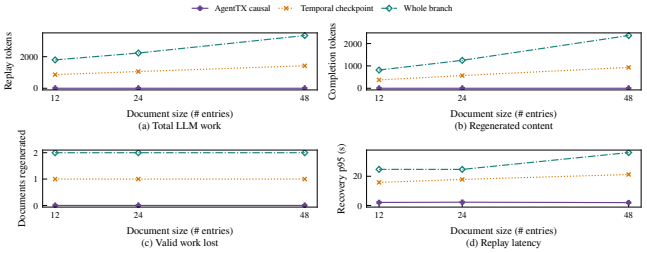


Figure 10: **Retained artifacts avoid controlled LLM replay.** *At 48 entries, temporal recovery replays 1,424.7 tokens and whole abort replays 3,340.3; zero excludes diagnosis, tests, and pre-failure work.*

Official SWE-Bench and Terminal-Bench recovery. The application suite charges recovery on official instances rather than on synthetic GitHub sidecars. After the selected policy, an oracle applies the official gold patch or Terminal-Bench solution; a live model is not required for this cell. The joint predicate still requires the official verifier *and* retention of independent notes.

Figure 11 reports all 18 cells (six tasks $\times$ three policies). Causal recovery satisfies both predicates on every task (6/6). Temporal recovery and whole-session abort pass the official tests after oracle repair but retain 0% of independent notes (0/6 each). The official verifier therefore measures task solvability, whereas independent-note retention exposes recovery granularity. Coarse policies pass the task after repair but discard valid suffix work that the verifier does not score.

7 Discussion and Limitations

Why not make causal rollback the default? The active view uses OverlayFS `index=on`, which preserves hard-link visibility within the tested session [36]. Recovery expands verified pre-existing and upper-created alias groups before reconstruction. Causal recovery still fails closed for arbitrary multi-object topology, bind mounts, external aliases, and unsupported generation changes.

Why trace instead of infer commands? Command strings are not a reliable specification of transitive compiler, test, or script reads. System-call tracing observes actual ac-

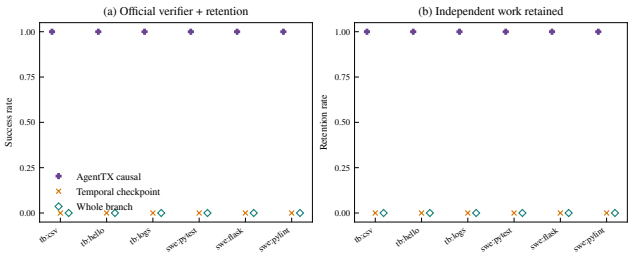


Figure 11: **Official workloads separate recovery from task repair.** *SWE-Bench Lite and Terminal-Bench, one oracle-repair run per cell. Causal recovery retains independent notes on 6/6 tasks; temporal and whole-session abort retain none, while still passing the official verifier.*

cesses but adds workload-dependent cost and remains Linux-specific. We therefore use the strace path for the main correctness and performance claims. In the measured 12-step comparison, it costs 25.60 ms/step versus 61.39 ms/step for the optional persistent eBPF backend. Both capture the same read and negative-lookup cases in this probe.

The eBPF path remains a replaceable implementation, not a claim of lower overhead. Neither tracer removes the need to define alias and unsupported-syscall semantics or repairs an inode relation already split by OverlayFS copy-up.

What state is outside the transaction? The current AgentTX prototype places workspace-local filesystem effects and ledger-bound tool-call history inside its transaction state. Hidden provider reasoning, network requests, emails, package registries, cloud control planes, process memory, and terminal state remain outside the guarantee. These effects may be irreversible or require subsystem-specific compensation; systems that capture full process or session state are complementary to AGENTTX.

What is outside the recovery scope? AgentTX determines and reconstructs the surviving speculative state. Publishing that state to a user workspace, coordinating concurrent publishers, and making publication crash-consistent are separate concerns outside this paper.

How strong is the current evidence? The native audit covers 20 successful trajectories and 638 turns, but only six turns pass the independence gate. The real-turn result is therefore a narrow counterfactual, not an end-to-end success study. Controlled motivation traces reach 128 tool-call steps, while failure/reload tests reach 256 steps. The matched strategy study uses 30 Codex cells and 25 native-SWEAgent cells per policy. It uses v8 for no rollback and retained v7 measurements for the other policies. The study therefore supports within-backend comparisons rather than paired protocol-version or cross-backend claims.

Application evaluation uses six SWE-Bench Lite and Terminal-Bench tasks with one oracle-repair run per policy. Live-agent control-plane runs remain seeded and use one model.

`try`, BranchFS, YoloFS, and Waypoint run artifact-natively on all 400 motivation manifests. `try` and BranchFS exercise native whole-unit discard; YoloFS and Waypoint exercise snapshot restoration. Waypoint runs in a privileged container, so these measurements do not support cross-environment latency claims. Sandlock builds on this host, but the kernel does not enable Landlock. Systems without a reusable recovery artifact are represented only by optimistic boundary semantics. We do not attribute those proxies to an external implementation.

8 Related Work

AgentTX is closest to effect-capturing semisolates, agent-oriented filesystems, and checkpoint or branch runtimes. Its distinguishing recovery unit is a dependency-defined effect subgraph within one speculative trajectory. We compare that semantic choice using matched recovered states, while keeping runtime claims separate across host and container environments.

Effect capture and semisolates. `try` introduces unprivileged semisolates for opaque commands, including effect introspection, optional application, and stacking [5]. Sandlock confines a single process run and makes its filesystem effects reversible through a copy-on-write work directory [31]. AGENTTX uses `try`'s mechanisms as its execution substrate. AgentTX extends this substrate from isolated calls to one speculative trajectory, then recovers a dependency-defined effect subgraph through historical reconstruction and context-aligned rollback.

Agent-native filesystems and branches. YoloFS stages mutations, exposes snapshots, and uses progressive permission to improve agent/user control [6]. Its filesystem mediation and permission interface are broader than AGENTTX's current substrate. BranchFS provides nested copy-on-write branch contexts with atomic leaf commit/abort and first-commit-wins coordination [7]. These systems select snapshots or branches. AgentTX instead selects a dependency-defined effect subgraph inside one evolving trajectory.

Checkpoint and restore. Waypoint combines filesystem deltas with process, memory, shell, and terminal state for branchable sessions [8]. DeltaBox accelerates complete sandbox checkpoint/rollback using incremental filesystem and process state [9]. Crab observes turn-level effects to avoid checkpoints that do not contain recovery-relevant state [33]. Their state coverage or checkpoint efficiency is complementary to AGENTTX. AGENTTX instead answers which non-contiguous filesystem effects to remove after a failure has been identified.

Process checkpointing. CRIU and DMTCP preserve complete process/session state for restart, migration, or debugging [10, 11]; their recovery unit is an application image, not a selected filesystem effect subgraph. AgentTX instead keeps the existing unmodified application interface, infers effects at opaque tool boundaries, and preserves valid later work after a selective failure.

Filesystem topology and tracing. OverlayFS specifies

copy-up, whiteout, and index semantics that determine when two pathnames still denote one object [36]. System-call tracers such as `strace` observe accesses made by arbitrary descendants [34]. The resulting paths still need hierarchy, descriptor, and alias normalization. Our contribution connects that observation layer to a dependency ledger and a fail-closed reconstruction policy. A different tracer alone does not solve object identity or selective rollback.

The surrounding systems literature exposes complementary boundaries. MicroVM and userspace-kernel sandboxes study isolation cost [37, 38], while provenance systems record whole-system data and control flow [39]. Kernel tracing work uses eBPF for low-overhead event capture and distributed path reconstruction [40, 41]. These mechanisms can improve observation, but they do not select the rollback set or align model context with the reconstructed workspace.

9 Conclusion

Long-running coding agents need a speculation unit larger than one command and a recovery unit smaller than one session. AGENTTX keeps the trajectory speculative while recovering a dependency-defined subgraph of versioned effects. It combines effect capture, object-aware history, selective workspace reconstruction, and context-aligned rollback.

Across 144 controlled runs, it retained all independent work and removed all invalid descendants. At 48 document entries, it avoided 1,424.7 replay tokens over temporal recovery and 3,340.3 over whole-session abort. These claims cover supported workspace-local filesystem effects with fail-closed object identity. The benefit scales with dependency sparsity rather than arising in every turn.

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